MADHAV RAMESH

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Education

University at Buffalo, The State University of New York — Buffalo, NY
Master of Science in Electrical Engineering — Expected January 2027
GPA: 3.75/4.00 | Relevant Coursework: Power Electronics, Power Distribution Systems, Photovoltaic Tech and Applications, Industrial Control Systems, Optimization of Next-Gen Smart Grid using AI

- Graduate student in Electrical Engineering with a focus on Power Electronics and Energy Systems.
- Seeking an opportunity to apply technical expertise in renewable energy, electric vehicles, and power conversion systems.

TKM College of Engineering — Kollam, India
Bachelor of Technology in Electronics and Communication Engineering — July 2022
GPA: 8.23/10.00 | Relevant Coursework: Power Electronics and Instrumentation, Control Systems

- Graduated in Electronics and Communication Engineering; combined academic rigor with practical expertise.
- Motivated by a focus on automation and innovation, aiming to deliver impactful solutions in dynamic engineering environments.

Work Experience

Renewable Energy Intern — GMR Group, New Delhi, India
June 2026 – July 2026

- Conducted comparative renewable-energy analysis across 8 Indian states, evaluating policies, incentives, approval mechanisms, land provisions, and project-development requirements, for a solar-wind hybrid power plant.
- Assessed substations, grid connectivity, evacuation capacity, transmission constraints, and planned network expansions for connection feasibility.
- Compared open-access economics, including wheeling, intrastate transmission, CSS, additional surcharge, electricity duty, and other state-specific charges.
- Plotted the projected energy consumptions using MATLAB and included in final report.
- Collaborated with the team to identify opportunities and propose our findings.

Software Engineer — Quest Global, Trivandrum, India
June 2023 – June 2025

- Delivered world-class end-to-end engineering solutions by leveraging deep industry knowledge and digital expertise.
- Collaborated with cross-functional teams to address critical and large-scale challenges across various industries.

Software Engineer — Onesoft Technologies, Kochi, India
August 2022 – May 2023

- Designed and implemented innovative technology solutions to drive client business growth.
- Delivered high-quality web applications with a strong focus on client satisfaction and precision.
- Collaborated with diverse teams to enhance testing efficiency and system reliability.

Volunteer Experience

Volunteer — Kerala Flood Relief Fund, Trivandrum, India
August 2018 – August 2018

- Contributed by being a part of the Flood Relief operations in Kerala, India during the unprecedented floods of 2018.
- Collaborated with a group, helping in the distribution of goods at a relief camp for a few days.

Academic & Technical Projects

Identifying Infeasibility in AC Generation Dispatch — Optimization of Next-Gen Smart Grid using AI
January 2026 – May 2026

- Improved an existing model of a 30 bus, 6 generator system with 41 transmission lines in MATLAB.
- Implemented constraint relaxation and perturbed complementary slackness to identify critical buses.
- Collaborated with a group to implement the MATLAB code.

Enhancing Distribution Network Resilience via Network Reconfiguration under Extreme Weather Events — Power Distribution Systems
August 2025 – December 2025

- Focused on improving resilience of 33 bus distribution network by implementing network reconfiguration to withstand weather-induced outages.
- Collaborated with a group to optimize the network using Julia.

Technical Skills

Software: MATLAB, Helioscope, AutoCAD, Julia
Programming: JavaScript, Python, Java, C++
Technical: Basics of energy system modeling, load analysis, grid interconnection and electrical circuit design
Languages: English, Malayalam, Hindi